

ARNUR JUMABEKOV

Hong Kong | +7-705-143-2007 | arnurjumabekov@gmail.com
[linkedin.com/in/arnurjumabekov](https://www.linkedin.com/in/arnurjumabekov) | github.com/turbokazax | [Robotics Portfolio](#)

EDUCATION

The Chinese University of Hong Kong (CUHK)

BSc in Mathematics and Information Engineering — first-year undergraduate, Class of 2030

Aug 2026 – Jul 2030 (expected)
Hong Kong

- Dual-faculty program, 10–15 admits/year; CUHK Admission & Engineering Faculty Scholarships

Haileybury Almaty

High School Diploma (International A-Levels), 100% merit scholarship (≈\$80k)

Aug 2023 – Jun 2025
Almaty, Kazakhstan

- **A-Levels:** Further Mathematics (A*), Mathematics (A*), Computer Science (A*), Physics (A); coursework incl. probability & statistics
- **SAT:** 1550/1600 (Math 800) | **AP:** Calculus BC (5/5), Computer Science A (5/5)

COMPETITIONS & AWARDS

- **Regeneron ISEF 2025: Special Award** from Regeneron & Mawhiba (full KFUPM scholarship); **Finalist**, 1 of 3 from Kazakhstan
- **Asian Pacific Mathematics Olympiad (APMO): top 54 of 2,000+** in Kazakhstan (2023); competed (2022)
- **Bronze:** Republican Math Olympiad (2022), **top 20 of 7,000+** nationally · **International Zhautykov Olympiad** qualifier (2022–24)
- **Silver:** Iranian Geometry Olympiad (2024), **top 3%** nationally · **Silver:** Sharygin Geometry Olympiad (2022), **top 4%** nationally
- **Gold:** Republican Science Projects Fair (2025): **top 1 of 433** national finalists, ≈4,000 entrants; absolute 1st in Informatics section
- **Silver:** British Physics Olympiad R1 (2024) · **Bronze:** Fizmat AI Olympiad (2024), top 13% of 299 (national ML & competitive programming olympiad)

PUBLICATIONS

First author — Jumabekov, A., Yashtini, M., “WeCAViT: A Weighted CNN Model for Pneumonia Detection in Chest X-rays,” **IEEE ICICIP 2025**. ieeexplore.ieee.org/document/10898141 · Cited in IEEE ISCS 2025 and IEEE ICPEGE 2026

EXPERIENCE & LEADERSHIP

Research Assistant (gap year), Smart Pipe Inspection Robot — ARMS Lab, Nazarbayev University

Aug 2025 – May 2026

- Built the Python motor-control API and calculus-based path-following algorithms for the robot’s manipulator arm
- Developed a YOLOv8 deep-learning model for in-pipe defect detection (welds, porosity, corrosion) for future field deployment

Robot Programmer — FTC Teams #24784 “Nyx Pardus” & #21063 “Initium Robotics”

Aug 2023 – Jun 2025

- Won the **Think Award** at the FTC Cyprus Championship; qualified for the Central Asia FIRST Championship twice
- Developed modular Java robot-control code with OpenCV object detection; led team game strategy, mentored junior programmers

Student Researcher — supervised by Prof. Maryam Yashtini (Georgetown); Pioneer Academics

May 2024 – Oct 2024

- Learned and applied the full independent-research cycle: formulated a hypothesis, synthesized the computer-vision literature, designed and ran model-comparison experiments in PyTorch, and wrote a 20-page research draft
- Rated top 5% relative to all undergraduates the supervisor has ever mentored

Intern & Lead Programmer, Tensegrity Robot — ISSAI, Nazarbayev University

May 2024 – Jul 2024

- Cut actuator position error to under 5% on the Tensegrity Robot, an assistive platform for early cerebral-palsy intervention in children, by writing OOP C++ Arduino firmware and tuning closed-loop PID control; built a real-time Python pipeline that cross-checked onboard sensor readings against OptiTrack motion capture of the robot’s moving top platform

Mentoring & Teaching

2022 – 2025

- **Haileybury Math Olympiad Club** (2024–25): coached 5 students in geometry & number theory in weekly 90-min sessions; 3 of 5 won regional Euler Olympiad awards
- **FTC #23551 “Ultron”, Morocco** (2024): remotely mentored the team in Java robot programming; the team won the **Think Award** (1st, engineering portfolio) and **Inspire Award** (2nd, FIRST’s top honor) at Morocco Nationals
- **American Space, Almaty** (2022): founded an Android development course; built the curriculum, recruited and led 3 mentors, taught 15+ students to 100% completion and 6+ Google-Play-ready group projects

PROJECTS

The Retention Architect — HackNU 2026 hackathon, Higgsfield AI track · Python, CatBoost, Optuna, SHAP

Apr 2026

- Predicted 3-class churn from 14 days of behavior; placed top 15 of 60 teams. Engineered 217 features across 90,000 users, tuned CatBoost with 50-trial Optuna, and validated out-of-time on a 7,000-user blind set
- Translated SHAP attributions into five business reason buckets for retention interventions

WeCAViT — Python, PyTorch, HuggingFace Transformers, scikit-learn

Jun 2024 – May 2025

- Designed a weighted ensemble of CNNs (VGG-16, ResNet-50, MobileNetV3), CBAM attention, and a Vision Transformer in PyTorch; reached **93.59% accuracy and 95.02 F1** on chest-X-ray pneumonia detection (PneumoniaMNIST, 5,856 images)
- Outperformed ten SOTA baselines (CNNs, ViTs, hybrids) by ≈4–5% in accuracy and F1 across 17 ablation experiments
- Fine-tuned the model for COVID-19 and general lung-opacity detection, applying 5-fold cross-validation and Student’s t-test verification of results

QuantaSoil — Python, scikit-learn, OpenAI API, React

Sep 2023 – Nov 2023

- Built a crop-recommendation classifier (Random Forest, 99.32% accuracy vs. SVM/LogReg/k-NN on a 2,200-row Kaggle set); translated its predictions into human-readable advice with GPT-4 and shipped the pipeline as a React web app
- Won 1st place of ≈40 teams at the “AI × Climate Change” hackathon; selected into the ABC Incubator × NURIS startup program

SKILLS & INTERESTS

Mathematics: olympiad problem solving · proof writing · probability & statistics · quantitative analysis

Programming: Python, Java, C++, Kotlin, SQL · algorithms & data structures · Git, Jupyter, LaTeX, Claude Code

Machine learning & libraries: PyTorch, NumPy, scikit-learn, pandas, Matplotlib, OpenCV, HuggingFace Transformers, CatBoost, Optuna, SHAP

Interests: “Mafia” (social deduction), chess, Dota 2, music production · **Languages:** English (**IELTS 8.0**), Russian, Kazakh